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23 March 1978
SUPERSEDING
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MILITARY SPECIFICATION

COMPASSES, DRAFTING, BEAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers beam compasses, with micrometer adjustment, in cases.

1.2 Classification. Bars for the beam compasses shall be of the following types, classes, and sizes as specified (see 6.2).

Type I - One-piece bar.
  Class 1 - Aluminum, I or T beam bar.
  Class 2 - Wood beam bar or straight edge.
  Class 3 - Steel beam bar or straight edge.
  Size - 16-inch length.
  Size - 18-inch length.
  Size - 24-inch length.
  Size - 30-inch length.
  Size - 36-inch length.
  Size - 48-inch length.
  Size - 60-inch length.
  Size - 72-inch length.
Type II - Sectional bars.
  Class 1 - Aluminum.
  Class 3 - Steel.
  Class 4 - Nickel silver.
  Size - 30-inch length (assembled).

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
QQ-W-423	- Wire, Steel, Corrosion-Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-W-13518	- Wood Preservative: Tetrachlorophenol and Pentachlorophenol, Surface Sealing Compound.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 276 - Stainless and Heat-Resisting Steel Bars and Shapes.
B 122 - Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), and Copper-Nickel Alloy Plate, Sheet, Strip and Rolled Bar.
B 151 - Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Rod and Bar.
B 308 - Aluminum-Alloy Standard Structural Shapes Rolled or Extruded.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The beam compasses shall be of the micrometer type and shall consist of pen, pencil-lead, and needlepoint attachments, two clamps for holding various combinations of these attachments, the bar as specified (see 1.2), and the appropriate fitted case for each type.

3.2 First article (first produced beam compasses). The contractor shall furnish one or more beam compasses when specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce beam compasses that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Aluminum. Aluminum alloy shall conform to ASTM B 308, Temper T4 or T6.

3.3.2 Wood. Wood shall be kiln-dried, straight-grained hardwood, free from knots, checks, or other imperfections which could affect the serviceability of the compass.

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3.3.3 Steel. Steel shall comform to ASTM A 276, Type optional.

3.3.4 Nickel silver. Nickel silver shall conform to ASTM B 151, Alloy optional for bar material, and ASTM B 122, Alloy optional.

3.3.5 Steel wire. Steel wire for springs shall conform to QQ-W-423, Form I, Composition 302, Condition B.

3.4 Performance. The beam compass when extended to its full length shall be capable of drawing two 360-degree circles so that a gap of not more than 0.005 inch per foot of radial length is measurable between the two circles. One circle shall be drawn clockwise and the other circle drawn counter-clockwise retracing the original circle.

3.5 Attachments. Attachments shall be interchangeable for use with either of the clamps. The pen attachments shall provide for an integral pen of the spring-blade type. Pen blades, the adjusting nut, and the screw stud shall be of corrosion-resistant steel or nickel silver. The pencil attachment shall be provided with a clamp for holding the lead. The points on the needlepoint attachment shall be either integral with the attachment or of the replaceable type and shall be made of corrosion-resistant steel. The following attachments shall be furnished with the compass:

1 pen attachment.
1 pencil-lead attachment.
2 needlepoint attachments with divider points.
1 needlepoint attachment with pivot needlepoint.

When needlepoint attachments with replaceable points are furnished, a total of two needlepoint attachments with extra divider and extra pivot points as specified (see 6.2) shall be required in lieu of the three needlepoint attachments listed above. All springs and screws shall be of corrosion-resisting steel.

3.6 Lead holder. A tubular metal lead holder for holding spare needlepoints and leads shall be furnished with each set. At least three leads in assorted degrees of hardness shall be furnished with each compass.

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3.7 Bars. All bars shall be straight within a tolerance of 0.062-inch per foot of length. Variations from straightness at any point on the bar shall not cause binding or tightness that could prevent the compass from performing as required.

3.7.1 Type 1. Bars for Type 1 beam compass shall be of cross-section design that will prevent attachment rotation within a tolerance of plus or minus 0.004 inch. The edges shall be smooth and free of burrs. When wood is specified (see 1.2) the bars shall be finished with varnish conforming to TT-V-121.

3.7.2 Type II. Bars for Type II beam compasses shall be in three sections joined by means of integral socket joints. Joints shall have mating splines and grooves or shall be shaped so that the bars will not rotate with relation to each other. Bars and clamps shall have mating grooves and splines, or the bars shall be so shaped that the clamps will not rotate with relation to the bar. The bar sections shall fit together sufficiently snug to comply with the performance requirements specified herein.

3.8 Clamps. Two clamps shall be provided with clamping screws, made of corrosion-resisting steel, for fastening the clamps in a fixed position on the bar. One clamp shall be provided with a micrometer-length screw adjustment for laying out and accurately measuring distance. The micrometer adjustment device shall provide a minimum of 0.250-inch linear travel for the point. Each clamp shall be provided with a socket and set screw for holding the attachments. Each clamp for the Type I, Class 2, beam compass shall be provided with a metal clamping strip to prevent the clamping screw from marring the surface of the bar. Clamps for Type I beam compasses shall be made of nickel silver, chrome-plated brass, or corrosion-resistant steel. Clamps for Type II beam compasses shall be made of the same material as the bar sections.

3.9 Cases. Unless otherwise specified (see 6.2) cases shall be wood, sanded smooth, and treated with wood preservative conforming to MIL-W-13518, Type I, Class 1. The cases shall then be finished with two coats of varnish conforming to TT-V-121, or shall be covered with leather, imitation leather, or plastic. The inside of the case shall be lined with dark-color felt or velvet cloth and shall provide a separate space or compartment for holding each component and attachment of the compass. Covering material and lining shall be of fungus-resistant material or shall be treated to resist fungus growth. The case shall have a hinged cover and a corrosion-resistant steel fastener.

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Cases for Type I beam compasses shall be of a size and shape for holding all the components of the beam compass, exclusive of the bar. Cases for Type II beam compasses shall be of a size and shape for holding all the components of the beam compass, including the sectional bars.

3.10 Identification marking. The beam compasses shall be identified in accordance with MIL-STD-130.

3.11 Workmanship. Bars, clamps, and attachments shall be free from burrs, cracks, and tool marks. All screws shall work freely without binding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced beam compass inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced beam compass inspection.

4.3.1 Examination. The first produced beam compass shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Test. The first produced beam compass shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The beam compasses shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Components or parts missing.
103. Attachments not as specified.
104. Lead holder not as specified.
105. Bars not as specified.
106. Clamps not as specified.
107. Case not as specified.
108. Identification marking missing, incomplete, or illegible.
109. Workmanship not as specified.

4.5.2 Performance test. The compass, extended to its full length and provided with a 6H lead sharpened to a point, shall draw a complete circle in a clockwise direction. From the same starting point the compass shall retrace the complete circle in a counter-clockwise direction. Inability of the compass to draw the complete circle clockwise without a discernable gap where the end of the circular line is joined to the starting point or inability to retrace the complete circle counter-clockwise without a gap or gaps exceeding 0.005 inch per foot of radial length between the two circles shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall constitute one defect.
111.	Compasses and bars not individually placed in boxes as specified.
112.	Box closure not Method V as specified.
113.	Strapping not as specified.
114.	Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each beam compass with all attachments, and sectional beam if applicable, shall be placed in the case. The case shall then be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. When a one piece bar is furnished, the bar shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.1 Compasses with Type I bar. Each preserved case, containing the compass and each preserved Type I bar, shall be placed individually in a close fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

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5.1.1.2 Compasses with Type II bar. Each preserved case, containing the compass and bar, shall be placed in a close fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Each complete beam compass with bar shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Complete beam compasses with bar, of like description, shall be packed in close fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A.

Note: If compasses required to be packaged Level A are shipped as a single item, the containers specified in 5.1 shall be used as the shipping container without additional over-packing.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A except that the boxes shall be Domestic type and strapping shall be Finish A.

5.2.3 Commercial. Beam compasses with bars shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Beam compasses are intended for use in drawing the arcs of large circles.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and size of bar to be furnished (see 1.2).
(c) Time frame required for submission of first produced beam compasses, and number of beam compasses when required (see 3.2).
(d) Number of extra pivot points required (see 3.5).
(e) Case material required, if other than wood (see 3.9).
(f) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(g) When Finish A strapping is required (see 5.2.1).

6.3 First produced beam compasses. Any changes or deviations of production beam compasses from the first produced beam compasses during production will be subject to the approval of the contracting officer. Approval of the first produced beam compasses will not relieve the contractor of his obligation to furnish beam compasses conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Classification changes. Changes in classification of the compasses between this revision of this specification and the previous edition have added two new classes. The additions are as follows:

Old Designation	New Designation
None	Type I, Class 3
None	Type II, Class 1

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA - GS

User activities:
Navy - MC, YD

Preparing activity:
Army - ME

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